



Data Driven Education

Natural Language Processing (NLP) Subteam
Presentation 2 - 3/11/2020

Five Sub-meetings: 2/7 2/14 2/21 2/28 3/6

Team Members



Yuntian Zhang

Project Manager
Data Scientist



Xinran Yu

Data Scientist



Morgan Powers

Project Manager
Web Developer

Researcher



Lindsey Blackmore

Web Developer

Researcher

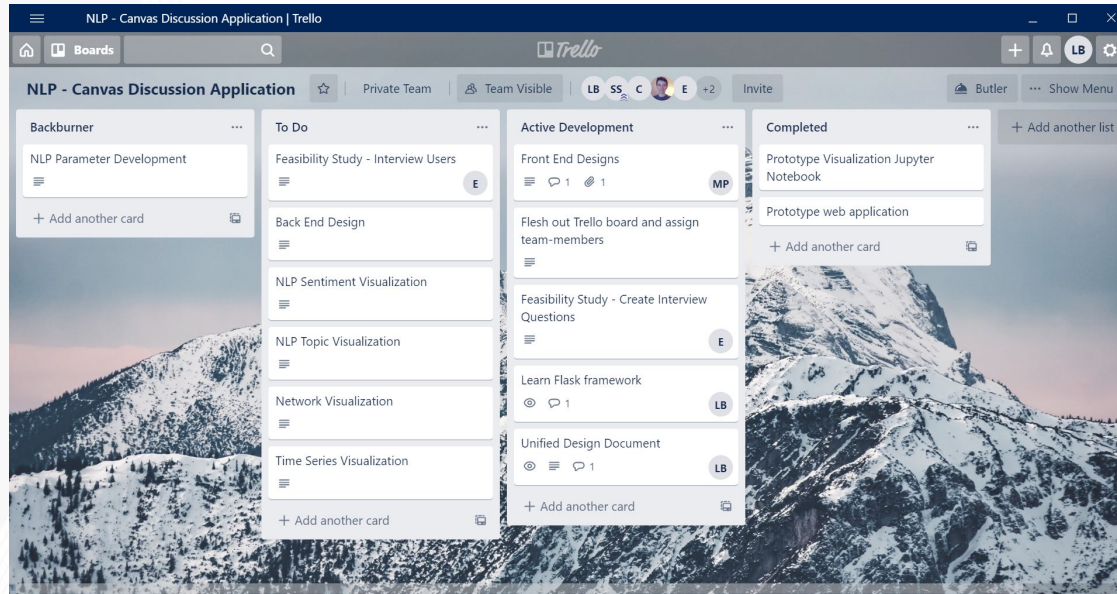


Erin Wrobel

Researcher

Overall Subteam Accomplishments

- Trello Board
- Consistent subteam meetings



Data Science Team

- edX data
 - 2 tables: forum_thread_ferpa & forum_comment_ferpa
 - 2667 comments in total: includes online classes such as CS1301, ISYE6501, etc
 - Focus on CS1301-Online - 2137 comments in total

comment_id	course_id	body	anonymous	anon
421c305ad0a31001dc6	course-v1:GTx+CS1301xll	Enter your answer without spaces! I was going crazy trying all	☐	
	course-v1:GTx+CS1301xll	SmartBook that is :)	☐	
15f3c305ad0a09000ea1	course-v1:GTx+CS1301xll	What is the answer for this question? I tried everything but no	☐	
2923b0abc0991000993	course-v1:GTx+CS1301xll	Passed with output printed:11211On the bubble111This line is ve	☐	
15f3c305ad0a09000ea1	course-v1:GTx+CS1301xll	I had the same problem with this question, I will try your advice	☐	
4390a52203095600057a	course-v1:GTx+CS1301xll	ah, I see. And for ' I do it the other way around: ord("")	☐	
0356c305ad0a040008f0	course-v1:GTx+CS1301xll	I knew in my bones it was just a typo - I just had the gut feelir	☐	
	course-v1:GTx+CS1301xll	I upgraded to 'PERSON' and the exam questions I had answer	☐	
4167acf2d083d0010c0	course-v1:GTx+ISYE6501x	I'm interested as well!	☐	
05945d63509fd00010b	course-v1:GTx+ISYE6501x	Hi, no I have not. Could you please provide the link to join the	☐	
4167acf2d083d0010c0	course-v1:GTx+ISYE6501x	I'd be interested in joining as well	☐	

Data Science

- Cognitive
 - Used the template
 - Develop
 - Goal: as

Phase	Descriptor	Indicator	Socio-cognitive process
0: Non-cognitive presence			
• comments that are not clear (e.g., picture and code w/o comment) • thank you and socializing comments • logistics, deadlines, technical Q&As			
1: Triggering Event			
• disagree/conflict with prior knowledge. "I think this is right/wrong." "I'm stuck." "I have no idea." "I am clueless." • focus on problem itself • shorter than 2 sentences in general – just the question, no elaboration • do not show they understand the concepts or ways that are needed to solve the given problem			
2: Exploration			
• more explorative, some hesitation, asking or giving suggestions, less authority ("I am not a TA, but ...") • show that they understand something, but don't understand something else – having partial understanding ("I understand this but not that") • introduce new ideas or suggestion ("Maybe try this?")			
3: Integration			
• give someone solution • use textbook references or other credible sources to help find solutions • "[their solution]. This works for me."			
4: Resolution			
• include elaboration on the solution; • whenever comment is definite/sure. • "tested code", "it worked" • comments and solutions directly from TAs/instructors			

Jarrison as a

Data Science Team

- Building Machine Learning Model
 - try to categorize the cognitive presence
 - in the process of preparing train-set
 - Time-Line: Finished the 300 training-set & 100 testing-set before spring break

“I didn't print the quotes and it passed. I did put my print statements right after the setter definition(s) line. Maybe that is what the grader is looking for? I also used only 1 return statement per setter method and remmed out my testing code at the bottom. I would try that first. ” —Phase II: Exploration

“For me, it seems the test code at the bottom was interfering with output. When I commented that out, then it passed. ” —Phase III: Integration

Data Science Team

- Building Machine Learning Model
 - try to categorize the cognitive presence
 - in the process of preparing train-set
 - Time-Line: Finished the train-set 300 before spring break
- Combine model with basic data visualization/analysis (Future job)
 - basic natural language processing: sentimental/semantic analysis
 - bar chart/plot chart

Web Application Development Team

- Gained familiarity with Flask framework and Canvas API
 - Integrated sample python script which invokes the Canvas API
 - Displays the sentiment and polarity scores of each discussion post
- Completed mockups for two organization flows
- Began a unified design document

Data Dashboard Mockups

- Adobe XD Demo

Design Document

Software Design Document

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Section 1 - Project Description

1.1 Project

Canvas Discussion NLP Analysis Web Application

1.2 Description

A web application LTI which will aid instructors in interpreting and analyzing student discussion discourses within the Canvas Discussion feature. Measures will include: sentiment, topic, cognitive presence, and keyword analysis as well as interaction statistics.

Section 2 - Overview

2.1 Purpose

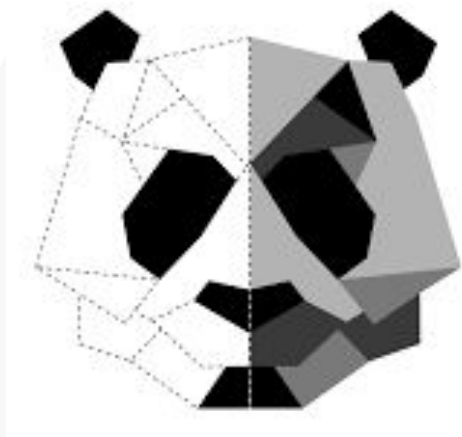
With possible FERPA violations with the use of Piazza in academic settings, this tool will aim to shift student discussion to the Canvas platform. Instructors will be able to monitor student discourse as well as analyze aspects of the discussions that are indicators of understanding and engagement.

2.2 Scope

It is the hope that this tool will be widely used by Georgia Tech instructors that employ a discussion forum for their courses. Since the web application will become an LTI, the tool could be used by any instructor that uses the Canvas platform.

Web Application Development Team

- Gain familiarity with Instructure UI component library
- Begin development once results from survey are received
- Continue thorough documentation of the web application



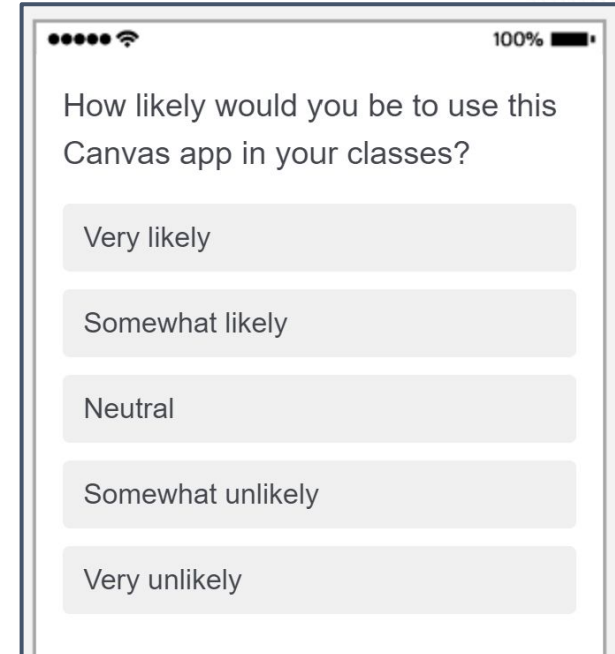
Education Research: Gathering Opinions

- **Goal:** determine usefulness and possible improvements for Canvas web app
- **Audience:** Professors and TAs in CS, Psychology, Math, and other disciplines
- **Method:** Qualtrics survey with app mockups

Education Research: Survey Summary

Specific areas addressed:

- Usefulness of product
- Navigable versus waterfall format
- Desired data
 - Sentiment analysis
 - Topic analysis
 - Student engagement statistics
 - etc.
- Most important factors
 - Interpretability
 - Aesthetic
 - Availability to students
 - etc.



How likely would you be to use this Canvas app in your classes?

Very likely

Somewhat likely

Neutral

Somewhat unlikely

Very unlikely